Anthropology Foundation (2025-26)

Handout 92: Neolithic Era (Part 1)

Contents

UPSC CSE PYQs

Causes, features, consequences of Neolithization

- **Examine Gordon Childe's** statement, "Neolithic culture is a **revolution**". (2010)
- Give an account of the **consequences of food production** of Neolithic culture. (2018)
- **Describe** the **features** of early farming cultures and Neolithic of the **Near East**. (2021)
- What are the **major theories** proposed in support of the **origin of food production**? How the change in subsistence economy brought **revolution** during this period? (2025)

What is Neolithic?

The **Neolithic culture** represents the final stage of the Stone Age, roughly dating from **10,000 BCE to 3,000 BCE** (with regional variations). The term Neolithic was first introduced by **Sir John Lubbock** in 1865. It is characterized by a **profound shift in human lifestyle** from mobile hunting-gathering to settled farming communities.

- Neolithic stage is associated with the **emergence of agriculture and domestication of animals.**
  - It fundamentally altered the human relationship with nature. For the first time in human history, communities began to rely substantially on **food production** rather than the direct appropriation of wild resources. Gradually, this shift led to **food surplus.**
  - In India, evidence from Mehergarh (Baluchistan) shows cultivation of barley and wheat around 7000 BCE and domestication of cattle, sheep, and goats.
- However, **hunting and gathering did not disappear** with the emergence of farming. What truly distinguishes the Neolithic is not the replacement of one subsistence strategy by another, but a broader **reorganisation of life**. The Neolithic should be understood as a complex transformation in human lifeways, marked by both continuity and change, rather than a simple agricultural revolution.

How did humans come to manage plants and animals over time?

The management of plants and animals did not emerge suddenly or through a single moment of invention. Archaeological evidence shows that domestication was a **long, gradual, and uneven process**, unfolding over thousands of years.

In this context, crucial analytical distinction must be made between **plant and animal collection/keeping** and **domestication**. The transition from collection to domestication becomes archaeologically visible through evidence from **archaeo-botany** and **archaeo-zoology**.

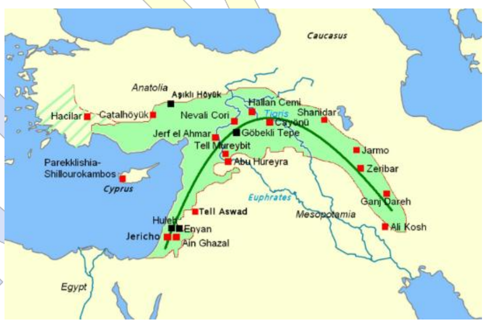
- Human practices unintentionally favoured traits advantageous to human use. Over successive generations, these selective pressures reshaped plant and animal populations.
- In **plants**, domestication is identified through **morphological changes** such as increased seed size, thicker seed coats, and the development of non-shattering rachis in cereals.
- In **animals**, domestication is inferred from changes in body size, horn shape, skeletal proportions, and age–sex profiles that reflect controlled breeding and selective culling.

Theory	Argument	Limitations
Climatic / Oasis Theory (V. Gordon Childe)	• Post-glacial desiccation forced humans, plants, and animals into oases, encouraging domestication.	• Overly deterministic, ignores human agency • Incorrect chronology • Lacks archaeological evidence of crisis.
Nuclear Zone Model (Robert Braidwood)	• Agriculture arose in natural habitats of wild progenitors where humans had long familiarity with species. • Hilly Flanks (Zagros–Taurus), Jarmo	• Explains *where*, not *why then* • Knowledge ≠ adoption • Weak on causal triggers.
Demographic Pressure Model (Lewis Binford)	• Population growth in core areas pushed groups into marginal zones, where domestication became necessary.	• Circular logic (pressure inferred from farming) • Questionable idea of "overpopulation", • No clear evidence of large-scale migration • Early sites show abundance, not scarcity.
Broad Spectrum Revolution (Kent Flannery)	• Late Pleistocene foragers broadened subsistence base • Intensification increased human–species interaction • Pre-adapting societies to agriculture.	• Explains *how*, not ultimate *why* • Trigger for broad-spectrum shift unclear
Co-evolutionary / Niche Construction (David Rindos)	• Domestication emerged unintentionally through long-term human modification of environments (niche construction)	• Downplays conscious choice and social motivation. • Treats domestication as largely unconscious

Social / Competitive Models (Brian Hayden)	• Agriculture developed to produce surplus for feasting, prestige, and social competition. It about social power rather than survival. • Gobekli Tepe, Catalhoyuk	• Assumes pre-existing inequality • Difficult to find direct evidence of early feasting • Not universally applicable
Symbolic / Cognitive Model (Jacques Cauvin)	• Change in worldview and symbolism preceded economic transformation. • Humans reimagined their relationship with nature first.	• Highly abstract, difficult to test • Chronology debated.

No single explanation adequately accounts for the Neolithic transition. What needs emphasis is the fact that domestication was a process which took considerable time and different trajectories. Today, most archaeologists see the beginning of food domestication as a **multicausal and region-specific process.**

Beginning and Spread



For a long time, archaeological thinking assumed that the Neolithic originated in a single core area and then spread to the rest of the world. This view is described as the **nuclear zone or diffusionist model**. Other regions were seen largely as recipients of this innovation.

This nuclear zone was identified as the zone in Near East (especially the Fertile Crescent). Here, a long sequence can be traced from Late Epi-palaeolithic hunter-gatherers (Natufian culture) to fully developed Neolithic societies. Archaeo-botanical evidence points to the domestication of a set of "**founder crops**" including wheat, barley, and legumes, while sheep, goats, cattle, and pigs were gradually brought under human management.

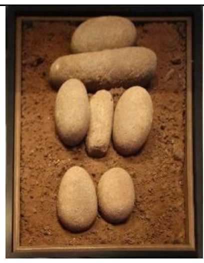
Levantine Sequence (Israel, Jordan, Lebanon, Syria)

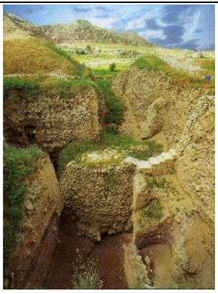
Period	Phase	Characteristic
c. 15,000 - 12,800 BCE	Early Natufian	**Semi-sedentary society** • Increase in human population • Round stone houses, lime-plastered floors, storage pits • Sickles for wild cereals (barley, wheat), • Heavy ground tools (mortars, querns) • Hunting of gazelle, small game • Burials with ornaments →social complexity and early symbolic life.
c. 12,800 – 11,700 BCE	Late Natufian	**Disruption due to Younger Dryas** • Many large Natufian villages were **abandoned.** Populations became more **mobile again.** Intensified wild cereal use persisted at core sites like Hayonim, but marginal areas saw mobility
c. 11,700 –	**PPN-A**	**Holocene warming:** Sedentism + early domestication. • Broad-spectrum foraging

11,000 BCE	(Experimental phase)	• Early **cultivation experiments** • Domesticated dogs • Mixed economy with still heavy reliance of hunting • First permanent villages with round houses, walls and towers, community structures (e.g. Jericho)

c. 10,800 – 8500 BCE	**PPN-B**	• **Full sedentism emerged** with multi-room rectangular houses in large villages
		• **Fully domesticated** cereals/pulses, and sheep/goats
		    ○ Emmer and einkorn wheat, barley in Levant
		    ○ Herding of Goat in Ganj Dareh and Ali Kosh (Zagros Mountain), Cattle domestication in upper Euphrates
		• **Multi-crop farming** is now primary activity.
		• Large and dense village, with clear property and household organization
		• Monumental architecture (Göbekli Tepe)
		• Elaborate ritual practices (plastered skulls)

c. 8500 onwards	**Pottery Neolithic**	• **Introduction of pottery** • Further population growth • Regional diversification of Neolithic cultures

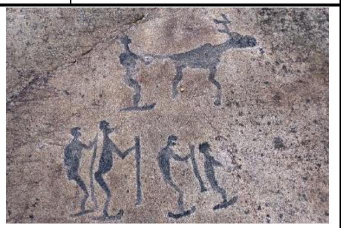
 Natufian mortars, grinding stones	 Stone face, Natufian culture	 Natufian Flutes from Eynan-Mallah
 Natufian Burials	 Natufian burial	 Schematic human figure made of pebbles, from Eynan, Early Natufian, 12,000 BC

 Tower of Jericho	 Catalhoyuk (Central Turkey)
 Ritual complex, Gobekli Tepe (Turkey)	 Jarmo (Iraqi Kurdistan, Zagros Mountains)

Diffusion to Europe

The Neolithic reached Europe from the Near East around 7000 BC. Its spread was uneven. Different cultural traditions took shape and moved across the continent over the next few millennia.

Pathways	6500-6000 BCE	6000-5000 BCE	5000-4000 BCE	4000-3000 BCE
Mediterranean Route	Sesklo culture (Early Greek Neolithic)	Impressa Ware culture (Italy)	Cardial Ware culture (Western Med.)	Atlantic Neolithic (France, Iberia)
Continental Route	Starčevo-Körös–Criș culture (Balkans)	Linear Pottery Culture (LBK) (Central Europe)	Post-LBK (Rössen, Lengyel) (Central Europe)	Funnelbeaker culture (N. Europe)

 Stonehenge, England	 Neolithic Orkney, Scotland	 Petroglyphs of Lake Onega and White Sea, Russia

Diffusion or Multiple Origin?

Initially, scholars believed that agriculture first emerged in the "nuclear areas" of the Fertile Crescent and then spread outward to other parts of the world through migration or cultural exchange. However, with advances in archaeology, genetics, and paleo-botany, discoveries in other regions revealed that agriculture developed independently in several parts of the world.

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- **China** witnessed at least **two distinct centres** of domestication: **millet** cultivation in the Yellow River basin and **rice** cultivation in the Yangtze basin.
    - Notably, pottery and sedentism appear early in East Asia, sometimes **before agriculture**, indicating a different sequence of change from West Asia.
- **Indian subcontinent**, particularly the north-western region represented by sites such as Mehrgarh, shows early cultivation of **wheat** and **barley** alongside pastoralism, while other regions later developed **rice**- and **millet**-based Neolithic cultures. These developments were shaped by local ecological and cultural conditions rather than simple diffusion.
- **Africa** saw independent domestication of crops such as **sorghum** and **pearl millet**.
- **The Americas** developed Neolithic transitions centred on **maize** in Mesoamerica and **potatoes** and **quinoa** in the Andes, with domestication processes unfolding slowly and unevenly, often without early large-scale sedentism.

Thus, the current consensus is that agriculture arose independently in at least 10–12 regions worldwide, with the Fertile Crescent being just one of several important centres. Thus, agriculture was not a single "invention" that spread, but a **global process that unfolded independently in several world regions.**

How did Labour and Technology Change?

This new relationship with nature carried significant implications for everyday human practice.

Domestication did not simply add new food sources; it demanded **regular, repetitive, and time-bound labour**. The growing dependence on managed resources necessitated new **technological responses**. Tools for clearing land, processing grain, storing surplus, and maintaining domesticated animals acquired greater importance.

- Tools became heavy, ground, and polished.
    - Because most of these were made from large stone pieces and were heavy, they are collectively known as **"heavy-duty tools."**
- **Common tools include:**
    - Polished stone axes/celt and adze (forest clearance and woodworking)
    - Chisels, hoes, sickles. Ring stone (agricultural tools)
    - Grinding stones, querns, pestles (grain processing)
    - Arrowheads, spear points, harpoons, and fishhooks (hunting and fishing)
    - Arrowheads, Mace heads, ring-stones, and stone balls (weapons)

- **Other Tools:**
    - **Microliths and blades** from earlier times continued in use.
    - Tools made of **bone and antler**, such as **needles, points, and arrowheads**.
    - **Decorative tools:** Beads, pendants, and figurines made from bone, stone, or shell.
- **Tool Technology**
    - **Raw Material:** Stone (especially flint, chert, basalt), bone, antler, and wood.

- **Advancement:** The Neolithic period saw the introduction of **grinding and polishing techniques**. The technique involved four steps: Flaking, Pecking, Grinding and Polishing.
- **Features:**
    - **Standardization:** Shapes were more uniform, reflecting specialized production.
    - The new method also allowed **resharpening and reuse**, freeing the craftsperson from limitations of breakage.

Other Technologies:

- **Pottery**
    - **A vast range of forms**—jars for storage, bowls for eating, cooking pots, and plates.
- Another invention was **weaving**.

How was Human Life Evolved?

- **Population increase:**
    - Long-term sedentism created a stable relationship between humans and their environment. Thus, gradually, human population size expanded.
    - However, this demographic expansion was accompanied by significant biological costs. There was a general **decline in average health** in early farming communities.
- **Sedentism (Settlements, Habitation)**
    - Intensified, time-bound work (e.g. cultivation and animal management) required prolonged presence in particular locations. As a result, there emerged **sedentism**. **Permanent habitation structure**, therefore, is linked with the neolithic economy.
    - Humans thus began living in **settled villages**, with some **element of planning.**
- Neolithic transformation marked the **first sustained human intervention in natural ecosystems.** (deforestation, soil management etc.)
- **Economy and Trade**
    - **Property:** Intensified labour and the need to manage land, storage, and herds encouraged the emergence of **property relations**.
    - Surpluses enabled **barter** and **limited trade**: In India, shells and semi-precious stones were traded between sites like Bagor and Balathal.
    - **Exchange of ideas and techniques** also occurred through **networks**.
- **Social Organization**
    - Over time, sedentary life and emergence of property laid the foundations for **division of labour, social differentiation and inequality**, even within relatively small village communities.
    - **Labour within households** became increasingly **structured and differentiated**, often along age and gender lines.
    - Communities likely had **kinship**-based social structures, with early signs of **leadership**.
    - People devised new ways and **norms of interaction and co-operation**.
- **Religion**
    - Changes in subsistence practices would have involved **shifts in symbolic and belief systems**.
    - **Figurines**, especially of fertility or mother goddesses, are found worldwide: Catalhoyuk (Turkey), Mehrgarh etc.

  - Archaeological evidence shows the **emergence of purposefully structured burials**, sometimes placed within or near domestic spaces.
    - **Such burials** do not appear for the first time in the neolithic phase, but they do increase in number.
- **Art and Craft**
  - **Pottery** often decorated/painted with geometric or symbolic motifs.
  - **Ornamentation** also became more prominent. They reflect emerging aesthetic traditions and social signalling.
  - At sites such as **Mehergarh**, tools and ornaments made from stone, bone, and antler indicate specialized craftsmanship.

Thus, this transformation was not limited to only subsistence production and the modification of landscape but also witnessed changes from the most material to the most symbolic - habitat, technology, demography, social organization, settlement and the use of space, and artistic and religious expressions were equally involved. Ultimately, it led to the emergence of first civilizations of the world.